

Ques. HOW DOES OPTICAL CHARACTER RECOGNITION WORK?**Deepak Negi**

Ans. Optical character recognition, or OCR, is a method of converting a scanned image into text. When a page is scanned, it is typically stored as a bit-mapped file in tagged image file (TIF) format. When the image is displayed on the screen, we can read it. But to the computer, it is just a series of black-and-white dots.

Follow the steps to understand OCR:

- When a scanner reads the image of a document, it converts the dark elements (text and graphics) on the page to a bitmap, which is a matrix of square pixels that are either on (black) or off (white).
- OCR software reads the bitmap that the scanner created and averages out the zones of on and off pixels on the page, in effect mapping the white space.
- In its first pass at converting images to text, the software tries to match each character through a pixel-by-pixel comparison to character templates that the program holds in memory.
- Characters that remain unrecognised go through a more intensive and time-consuming process called feature extraction.
- But, these two processes do not decipher every character; OCR programs take two approaches to the remaining hieroglyphics. Some other OCR programs invoke a specialised spelling checker to search for errors and correct these. For example, to OCR programs, number 1 and letter l look very similar, so do 5 and S, or cl and d.
- Most OCR programs give you the option of saving the converted document to an ASCII file or in a file

format that popular word processors can recognise.

Q2. WHY DO AUTOMOTIVE INDUSTRIES PREFER CAN OVER SPI OR I2C?**Ramakanta Mohanta**

A2. Serial peripheral interface (SPI) and inter-integrated circuit (I2C) are protocols for serial communication among devices connected to the same bus and communicating serially in master-slave mode. In SPI, three or more signal wires are needed (clock, data in, data out and multiple slave selects). Number of devices in automobile is large and therefore slave select lines will be more. The system is wire-intensive, directly proportional to the number of slaves, for application in SPI protocol. This makes SPI unsuitable for applications in automobiles. So comparison can only be made between CAN and I2C.

I2C is a synchronous serial communication on two signal wires. Communication is node based. It was developed by Phillips for application in their products in the entertainment field. Data is exchanged between devices that most of the time are on the same card. Distance is a limitation in the protocol.

Controller area network (CAN), on the other hand, is a message based protocol for asynchronous serial bus communication. It was developed by Bosch in the early 1980s. It defines a standard for efficient and reliable communication between the sensor, actuator, controller and other nodes in real-time applications.

In a high-end car, there are 50-plus dedicated computers for every subsystem like fuel control system, auto transmission control and brake control. Data between sensors and controllers is exchanged at a very high speed. So we need a sophisticated protocol to con-

nect all computers. CAN protocol, over a period of time, has developed to meet this requirement. It is now the *de facto* international standard for a large variety of networked embedded control systems.

Q3. CAN YOU EXPLAIN HOW TO DESIGN A SOLAR CHARGE CONTROLLER AND PROVIDE LINKS FOR THE SAME?**Vignesh Vicky**

A3. A solar charge controller monitors the state of charge of the battery bank, controls the charging process and connection/disconnection of loads. This optimises battery use and significantly extends its life. The solar charge controller may only be used in photovoltaic (PV) systems for charging and controlling lead-acid batteries in accordance with its operating manual and charging specifications of the battery manufacturer.

Please find below links for some projects based on a solar charge controller, published in EFY:

<http://electronicsforu.com/electronics-projects/hardware-diy/smart-solar-charge-controller>
<http://electronicsforu.com/electronics-projects/hardware-diy/microcontroller-based-solar-charge-controller>

In the solar-powered lighting system, the system's overall success depends mainly on the solar charge controller. It is an indispensable link between the solar panel, battery and load.

Main function of a solar charge controller is to limit the rate at which electric current is added to or drawn from batteries. It prevents over-charging and protects the battery from voltage fluctuations, which can reduce battery performance or lifespan.

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